NATIONAL UNIVERSITY OF MODERN LANGUAGE ISLAMABAD

Lab final

Submitted to:

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Task 01

Code;

```
import java.util.Scanner;

abstract class Shape {
    abstract double calculateArea();
}

// Square class
class Square extends Shape {
    private double sideLength;

    Square(double sideLength) {
        this.sideLength = sideLength;
    }

    @Override
    double calculateArea() {
        return sideLength * sideLength;
    }
}

class Rectangle extends Shape {
    private double length;
    private double width;

    Rectangle(double length, double width) {
        this.length = length;
        this.width = width;
    }

    @Override
    double calculateArea() {
        return length * width;
    }
}

// Triangle class
class Triangle extends Shape {
    private double baseLength;
    private double height;

    Triangle(double baseLength, double height) {
        this.baseLength = baseLength;
        this.height = height;
    }

    @Override
```

```java
    double calculateArea() {
        return 0.5 * baseLength * height;
    }
}

class ShapeFactory {
    public static Shape createShape(int choice, Scanner scanner) {
        switch (choice) {
            case 1:
                System.out.print("Enter the side length of the square: ");
                double side = scanner.nextDouble();
                return new Square(side);
            case 2:
                System.out.print("Enter the length of the rectangle: ");
                double length = scanner.nextDouble();
                System.out.print("Enter the width of the rectangle: ");
                double width = scanner.nextDouble();
                return new Rectangle(length, width);
            case 3:
                System.out.print("Enter the base length of the triangle: ");
                double base = scanner.nextDouble();
                System.out.print("Enter the height of the triangle: ");
                double height = scanner.nextDouble();
                return new Triangle(base, height);
            default:
                System.out.println("Invalid choice. Please select a valid
option.");
                return null;
        }
    }
}

public class ShapeAreaCalculator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("Select a shape to calculate area:");
        System.out.println("1. Square");
        System.out.println("2. Rectangle");
        System.out.println("3. Triangle");

        int choice = scanner.nextInt();
        Shape shape = ShapeFactory.createShape(choice, scanner);

        if (shape != null) {
            System.out.println("Area: " + shape.calculateArea());
        }

        scanner.close();
    }
```

```
}
```

Output:

```
PROBLEMS 8    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\chdaw\Desktop\java>  & 'C:\Program Files\Eclipse Adoptium\jdk-21
Roaming\Code\User\workspaceStorage\dcc627994f71fe0e709f8f7ae82f6c12\redhat.j
Select a shape to calculate area:
1. Square
2. Rectangle
3. Triangle
1
Enter the side length of the square: 2
Area: 4.0
PS C:\Users\chdaw\Desktop\java> 
```

Code Smells in the Program

1. **Switch Statement for Shape Selection:**
    - The use of a switch statement violates the Open-Closed Principle (OCP). Adding a new shape would require modifying the switch block, making the code less extensible.
2. **Mixed Responsibilities:**
    - The main method is overloaded with multiple responsibilities: shape selection, area calculation, and user interaction.
3. **Magic Numbers:**
    - The shape selection menu uses hardcoded numbers (1, 2, 3), which can make the code less intuitive.
4. **Duplicate Code:**
    - Code for user input and area calculation is repeated for each shape.
5. **Lack of Polymorphism:**
    - The design does not leverage object-oriented principles, such as polymorphism, to handle shape-specific behavior

Task 02

Part 1

```
public class Main {
    public static void main(String[] args) {
       int a=2;
       int b=0;
 try {
            int divide=a/b;
        } catch (ArithmeticException e) {
            System.out.println("Arithmetic Exception handled");
        }


}
}
```

Out put:

```
 and design\lab\final>
cd "d:\university\5-fifth semester\software construction and design
\lab\final\" ; if ($?) { javac Main.java } ; if ($?) { java Main }
Arithmetic Exception handled
PS D:\university\5-fifth semester\software construction and design\
lab\final>
```

Part 2

```
import java.util.InputMismatchException;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
 try {
            System.out.println("Enter an integer: ");
            int integer = sc.nextInt();
            System.out.println("You entered: " + integer);
        } catch (InputMismatchException e) {
            System.out.println("Error: Invalid input. Please enter an integer.");
        }
```

```
sc.close();
}
}
```

Output:

```
 and design\lab\final>                           > cd "d:\universit
y\5-fifth semester\software construction and design\lab\final\" ; i
f ($?) { javac Main.java } ; if ($?) { java Main }
Enter an integer:
asdf
Error: Invalid input. Please enter an integer.
PS D:\university\5-fifth semester\software construction and design\
lab\final>
```